

# Experiment 01: Overview of IoT Architecture & Protocols

## 5-Tier Architecture, Protocol Comparison & MQTT Telemetry Payload Code

### 1. Constrained IoT Application Protocols Matrix

| Protocol  | Transport    | Messaging Model         | Header Size     | Primary Application Domain                               |
|-----------|--------------|-------------------------|-----------------|----------------------------------------------------------|
| MQTT      | TCP (1883)   | Publish / Subscribe     | 2 Bytes (Fixed) | Real-time sensor telemetry, SCADA, battery-backed nodes. |
| CoAP      | UDP (5683)   | REST Request / Response | 4 Bytes (Fixed) | Constrained 8-bit MCUs, low-power lossy networks (LLNs). |
| HTTP/REST | TCP (80/443) | Client / Server Request | > 500 B (Text)  | Cloud-to-cloud APIs, enterprise REST integrations.       |
| WebSocket | TCP (80)     | Full-Duplex Stream      | 2–10 Bytes      | Live browser telemetry monitoring dashboards.            |

Table 1: Technical Comparison of IoT Application Layer Protocols.

### 2. Edge Microcontroller Telemetry Payload Firmware

Listing 1.1: Microcontroller Firmware for MQTT Telemetry Serialization.

```
1 /*
2  * Experiment 01: IoT Edge Telemetry Serialization & Protocol Ingestion
3  * Formats multi-sensor readings into lightweight JSON payloads for MQTT Broker publishing.
4  */
5
6 #include <ArduinoJson.h> // Standard lightweight embedded JSON library
7
8 void setup() {
9     Serial.begin(9600); // Initialize UART Serial Interface
10    Serial.println("=====");
11    Serial.println("   IoT Lab: MQTT Node Telemetry Payload Stream   ");
12    Serial.println("=====");
13 }
14
15 void loop() {
16     // Step 1: Acquire mock sensory inputs from Perception Layer
17     float currentTemp = 28.4;
18     float currentHumidity = 62.5;
19     int deviceUptime = millis() / 1000;
20
21     // Step 2: Construct serialized JSON payload packet
22     StaticJsonDocument<128> doc;
23     doc["node_id"] = "ESP32_NODE_01";
24     doc["uptime"] = deviceUptime;
25     doc["temp"] = currentTemp;
26     doc["hum"] = currentHumidity;
27     doc["status"] = "ONLINE";
28
29     // Step 3: Stream MQTT packet to Serial Gateway
30     Serial.print("TOPIC: sensors/room1/telemetry | PAYLOAD: ");
31     serializeJson(doc, Serial);
32     Serial.println();
33
34     delay(3000); // 3-second transmission cycle
35 }
```

### 3. Simulated MQTT Broker Telemetry Stream Output

```
=====
IoT Lab: MQTT Node Telemetry Payload Stream
=====
TOPIC: sensors/room1/telemetry | PAYLOAD: {"node_id":"ESP32_NODE_01","uptime":3,"temp":28.4,"hum":62.5,"status":"ONLINE"}
TOPIC: sensors/room1/telemetry | PAYLOAD: {"node_id":"ESP32_NODE_01","uptime":6,"temp":28.6,"hum":62.1,"status":"ONLINE"}
TOPIC: sensors/room1/telemetry | PAYLOAD: {"node_id":"ESP32_NODE_01","uptime":9,"temp":28.9,"hum":61.8,"status":"ONLINE"}
TOPIC: sensors/room1/telemetry | PAYLOAD:
{"node_id":"ESP32_NODE_01","uptime":12,"temp":29.1,"hum":61.5,"status":"ONLINE"}
```